State-of-the-Art Knowledge Distillation for Image Classification, 2022–2026: A Systematic Survey of Methods, Multi-Objective Optimization, and Reported Results

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Abstract

Knowledge distillation trains a compact student to imitate a larger teacher. This survey covers methods for image classification from 2022 to 2026. We screened the titles of 2,193 papers from CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, AAAI, TMLR, TPAMI, IJCV and arXiv, then 445 more found with broader title terms, and kept the 112 methods that meet three inclusion criteria set in advance. We organise them by what knowledge is transferred (logits, features and relations), how training is arranged, how far the teacher is trusted, and how the task and distillation objectives are combined, with a primer on multi-objective optimisation for the last group. We use multi-objective optimisation to organise these update rules; whether it explains their gains is open, since direct measurements of gradient conflict between the two objectives in distillation are scarce. For the commonly used CIFAR-100 and ImageNet-1K pairs we list the results each paper reports and flag protocol deviations. On CIFAR-100 ResNet32×4→ResNet8×4, where classical KD reaches 73.33%, the highest reported configurations are 79.25% (DeepKD, stacked on another method) and 78.76% (VRM, a single method), and most single logit methods sit between 76.0% and 77.8%. On ImageNet ResNet34→ResNet18 the spread is narrow: of the 79 methods that report this pair, all but seven score between 71.0% and 73.0%, against 71.70% for DKD. Many of the best results stack methods, change the teacher or report a single run, so small gaps do not support a ranking. We close with five open problems: a leaderboard re-run under one protocol that also tests what stacked results measure, which teacher signal to trust, how to combine the task and distillation objectives and whether their gradients conflict at all or only differ in scale, the gap between data-free, CIFAR-100 and ImageNet results, and metrics beyond top-1 accuracy.

1 Introduction

Knowledge distillation (KD) trains a small student network to match the outputs of a larger teacher, so that the student keeps much of the teacher's accuracy at a fraction of its inference cost (Hinton et al., 2015). The original recipe adds a temperature-softened Kullback–Leibler term to the cross-entropy loss. Logit methods split the KD loss into parts and reweight them (Zhao et al., 2022), standardise or rescale logits (Sun et al., 2024) or learn the temperature (Li et al., 2023c). Feature and relation methods align intermediate representations or the structure between samples (Tian et al., 2020; Chen et al., 2022; Zhang

et al., 2025). Other work changes the training scheme, asks when the teacher should be trusted at all, or changes how the task gradient and the distillation gradient are combined (Zhao et al., 2023; Hayder et al., 2026).

Motivation and gap. Most of this work reports results on the same few teacher–student pairs: CIFAR-100 with the teacher–student pairs of CRD (Tian et al., 2020) and ImageNet-1K with ResNet34→ResNet18. Yet numbers from different papers are rarely put side by side with their caveats, and recent gains are often within a point of each other. Table 1 compares this survey with existing ones: the broad KD surveys (Gou et al., 2021; Wang & Yoon, 2022) predate most methods covered here, and Mansourian et al. (2025) covers newer work. None of them states how its methods were selected, and none treats the combination of task and distillation objectives as a multi-objective optimisation problem in its own section. A survey of gradient-based multi-objective learning (Chen et al., 2026b) covers that side in depth but does not discuss distillation.

Scope and inclusion criteria. We survey state-of-the-art KD for image classification published from 2022 to 2026. We include a paper if it meets all three criteria below, which we set before screening began. **I1**: it proposes a KD method in which a student is trained with a teacher signal for image classification, in one of five families (logit or loss, feature or relation, training scheme, teacher reliability, objective combination). Analysis-only papers are cited as supporting work but not ranked. **I2**: it appeared between January 2022 and November 2026 at CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, AAAI, TMLR, TPAMI or IJCV, or is an arXiv preprint first posted from 2025 without such a venue. **I3**: it reports top-1 accuracy on CIFAR-100 with a standard CRD teacher–student pair, or on ImageNet-1K with ResNet34→ResNet18 or ResNet50→MobileNet. Excluded papers carry one of seven codes: E1 not KD, or KD only as a tool for another task; E2 not image classification; E3 venue or date outside I2; E4 no standard-protocol result; E5 survey or position paper; E6 dataset distillation or compression without a teacher; E7 withdrawn or duplicate version. Distillation for language models, vision–language models, detection and other tasks is outside the scope; Section 10 gives pointers.

Paper selection. We queried three groups of sources: the computer-vision venues; the machine-learning venues and TMLR; and AAAI, the two journals and arXiv, adding papers cited in recent comparison tables. This gave 2,193 records (656, 985 and 552), of which 310 were read in full. A second pass with broader title terms ("teach", "student", "mimic", "logit" and others) screened 445 further records, re-checked 29 cited works that none of the three queries had picked up, and added two methods. Fifteen papers that were first inaccessible were later retrieved by hand and added six more. The final set has 112 methods; two papers could not be assessed because no full text was found. An independent spot-check found no errors among the 20 sampled inclusions and 20 sampled exclusions whose full text was accessible (2 and 10 further sampled papers could not be read). The supplementary material gives the sources, queries, date and counts of each query group (Table S1), the per-record screening logs, and the complete list of the 112 included methods with each one's reported results and source table (Table S2); the main text cites only the methods it discusses (Fig. 1).

Taxonomy. Figure 2 shows how the methods are organised. The first axis is what knowledge is transferred: logits (Section 3) or features and relations (Section 4). The second is how training is arranged: offline, online, self-distillation, multi-teacher or data-free (Section 5). Two branches rarer in earlier surveys cover how far to trust the teacher (Section 6) and rules that combine the task and distillation gradients (Sections 7 and 8).

Contributions. The survey makes four contributions. First, it gives a method set for 2022–2026 selected with criteria set in advance and a logged decision for every paper, so a reader can check why a method is in or out. Second, it describes each family with its strengths, weaknesses and applicability. Third, it reviews multi-objective optimisation from the angle of distillation and places the KD methods that use it, including the priority-based PCD (Varam & AlHajri, 2026), among the general gradient-combination rules. We use this as a lens for organising update rules; whether the two gradients actually conflict in standard distillation is largely untested (§8.2). Fourth, it collects the reported CIFAR-100 and ImageNet numbers of the included methods in one place and flags the results whose setting differs from the standard protocol.

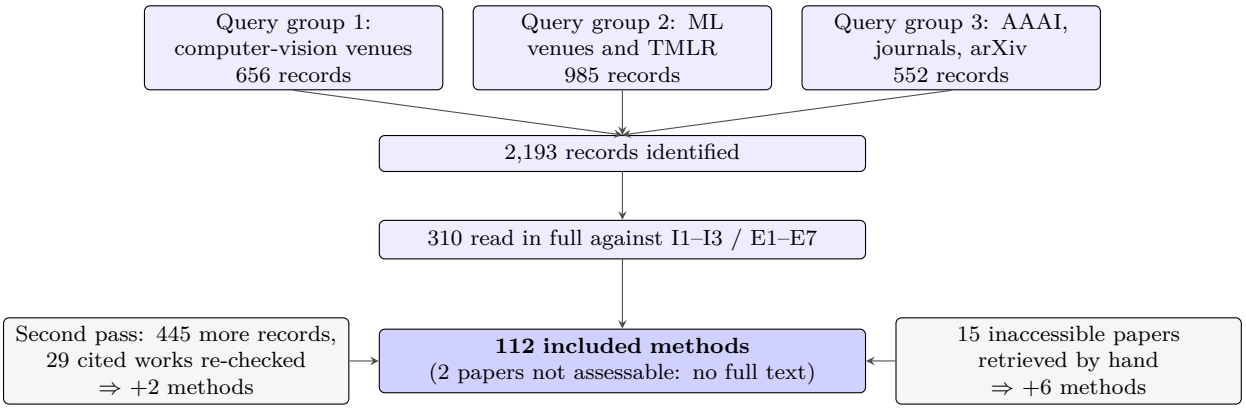


Figure 1: Paper selection. Counts are those of the three query groups, the second pass and the manual retrieval described in the text; I1–I3 and E1–E7 are the inclusion and exclusion criteria.

Section 9 lists the reported results, Section 10 discusses open challenges, and the supplementary material lists all included methods.

Table 1: Comparison with existing surveys. ✓: covered in a dedicated section; ○: covered partially or only in a narrower setting; ×: not covered. "MOO in KD": the distillation and task objectives treated with multi-objective optimisation (Pareto, gradient balancing or conflict handling); loss weighting alone counts as ○. "2024–26": discusses knowledge-distillation methods published in 2024–2026.

Survey	Venue	Year	Logit	Feat./ relation	Schemes	Teacher reliab.	MOO in KD	Bench./ results	LLM/ VLM	2024–26 methods
Gou et al.	IJCV	2021	✓	✓	✓	×	×	✓	×	×
Wang & Yoon	TPAMI	2022	✓	✓	✓	○	×	×	×	×
Mansourian et al.	TMLR	2025	✓	✓	✓	○	○	✓	✓	○
Chen et al.[†]	TMLR	2026b	×	×	×	×	×	×	×	×
This survey	–	2026	✓	✓	✓	✓	✓	✓	○	✓

[†]A survey of gradient-based multi-objective deep learning (first posted 2025): it covers MOO in depth but does not mention distillation.

2 Preliminaries

2.1 Notation and the classical objective

We consider C-way classification with an input $\mathbf{x}$ and a label $y \in \{1, \dots, C\}$. A pretrained teacher f_t and a student f_θ with parameters θ produce the logit vectors $\mathbf{z}_t = f_t(\mathbf{x})$ and $\mathbf{z}_s = f_\theta(\mathbf{x})$ in $\mathbb{R}^C$. In the standard (offline) setting the teacher is trained first and kept fixed while the student learns (Hinton et al., 2015). For a temperature $T > 0$ the softened class distribution is

$$p_c^T(\mathbf{z}) = \frac{\exp(z_c/T)}{\sum_{j=1}^{C} \exp(z_j/T)}, \qquad c = 1, \dots, C, \tag{1}$$

so $\mathbf{p}^1(\mathbf{z}_s)$ is the ordinary softmax. Hinton et al. (2015) train the student on the label and on the teacher's softened distribution at once; we write

$$\mathcal{L}(\theta) = \underbrace{\mathcal{L}_{\mathrm{CE}}\big(y, \mathbf{p}^1(\mathbf{z}_s)\big)}_{\text{task loss}} + \lambda \underbrace{T^2\, \mathrm{KL}\big(\mathbf{p}^T(\mathbf{z}_t) \,\big\|\, \mathbf{p}^T(\mathbf{z}_s)\big)}_{\mathcal{L}_{\mathrm{KD}}}, \tag{2}$$

where $\mathcal{L}_{\mathrm{CE}}(y, \mathbf{p}) = -\log p_y$ and $\lambda \geq 0$ sets how much the distillation term contributes. The convex form $(1-\alpha)\mathcal{L}_{\mathrm{CE}} + \alpha\mathcal{L}_{\mathrm{KD}}$ differs only by a global rescaling under plain SGD, but not exactly under Adam, weight

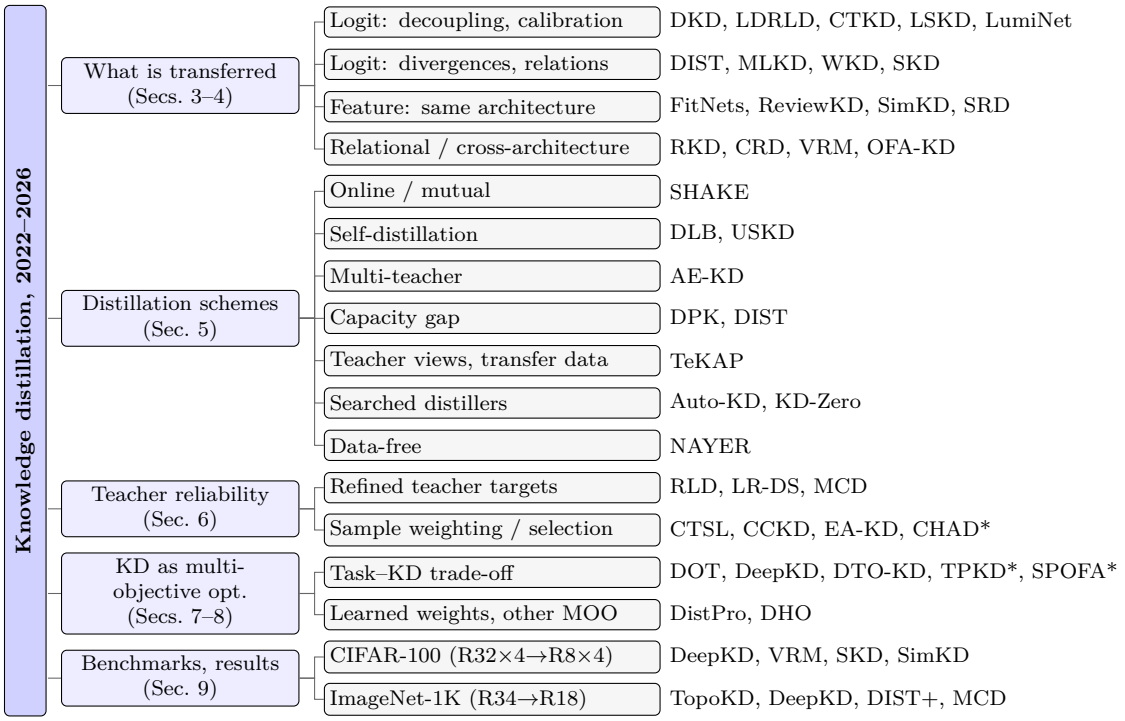


Figure 2: Taxonomy of the survey. Branches are the survey sections; each leaf lists representative state-of-the-art methods (2022–2026) discussed in that section. Methods marked * are preprints. Benchmark leaves list methods whose reported results anchor the comparison tables of Sec. 9. Labels point to the cited discussion in each section; Table S2 lists all 112 methods.

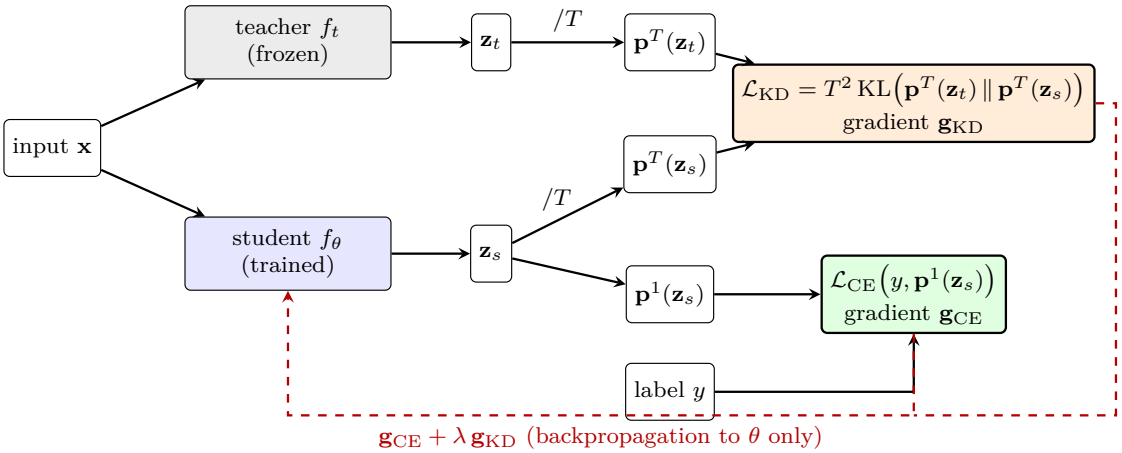


Figure 3: The classical teacher–student pipeline of Eq. (2). The frozen teacher and the student see the same input. The student's logits feed two losses: a cross-entropy against the label at $T = 1$ and a temperature-scaled KL divergence against the teacher. Only the student receives the combined gradient $\mathbf{g}_{\mathrm{CE}} + \lambda \mathbf{g}_{\mathrm{KD}}$ (Eq. (4)).

decay or gradient clipping. "KD" with no qualifier means Eq. (2), and the methods of §3 are described by what they change in $\mathcal{L}_{\mathrm{KD}}$, in $\mathbf{p}^T$, or in the weight λ. Figure 3 shows the resulting pipeline.

Per-objective gradients. Section 7 treats Eq. (2) as a two-objective problem. With respect to the student logits, the cross-entropy gives $\partial \mathcal{L}_{\mathrm{CE}}/\partial \mathbf{z}_s = \mathbf{p}^1(\mathbf{z}_s) - \mathbf{e}_y$, where $\mathbf{e}_y$ is the one-hot label, and Hinton et al. (2015) show that the softened KL term has per-logit gradient $(1/T)\big(p_c^T(\mathbf{z}_s) - p_c^T(\mathbf{z}_t)\big)$. After the T^2 factor this gives

$$\frac{\partial \mathcal{L}_{\mathrm{KD}}}{\partial \mathbf{z}_s} = T\big(\mathbf{p}^T(\mathbf{z}_s) - \mathbf{p}^T(\mathbf{z}_t)\big). \tag{3}$$

Writing $\mathbf{J}_\theta = \partial \mathbf{z}_s/\partial \theta$ for the student's Jacobian, the per-objective parameter gradients are

$$\mathbf{g}_{\mathrm{CE}} = \nabla_\theta \mathcal{L}_{\mathrm{CE}} = \mathbf{J}_\theta^\top \big(\mathbf{p}^1(\mathbf{z}_s) - \mathbf{e}_y\big), \qquad \mathbf{g}_{\mathrm{KD}} = \nabla_\theta \mathcal{L}_{\mathrm{KD}} = T\,\mathbf{J}_\theta^\top \big(\mathbf{p}^T(\mathbf{z}_s) - \mathbf{p}^T(\mathbf{z}_t)\big), \tag{4}$$

and plain gradient descent on Eq. (2) follows $\mathbf{g}_{\mathrm{CE}} + \lambda \mathbf{g}_{\mathrm{KD}}$ (in practice averaged over a mini-batch). We say the two objectives *conflict* on a batch when $\langle \mathbf{g}_{\mathrm{CE}}, \mathbf{g}_{\mathrm{KD}} \rangle < 0$, that is, when their cosine similarity is negative. This can happen even with a correct teacher: $\mathbf{g}_{\mathrm{CE}}$ pushes towards the one-hot vector, $\mathbf{g}_{\mathrm{KD}}$ towards a softened distribution with mass on non-target classes. Whether it matters in practice is the empirical question of §8.

2.2 The role of temperature

At $T = 1$ a confident teacher puts almost all its mass on one class and $\mathcal{L}_{\mathrm{KD}}$ adds little to the label. Raising T exposes the probabilities the teacher gives to wrong classes, which Hinton et al. (2015) call its knowledge of which classes resemble one another. Since soft-target gradients scale as $1/T^2$ at high temperature, Hinton et al. (2015) multiply by T^2 so that the two terms' relative contribution stays roughly unchanged as T is tuned, an approximation that holds in the high-T regime; in the high-T limit with zero-mean logits, KD reduces to squared-error logit matching (Hinton et al., 2015).

Three properties drive much of §3: T is one global hyper-parameter, usually grid-searched (Li et al., 2023c); sharing it asks the student to copy the teacher's logit *range*, not just its ranking (Sun et al., 2024); and the KD term mixes target and non-target knowledge, with target confidence scaling the latter down (Zhao et al., 2022). CTKD learns T, LSKD standardises the logits before the softmax, and DKD separates the two parts of the KL term; §3 compares them.

The KD temperature is different from the temperature used for calibration. Guo et al. (2017) fit one temperature post hoc so that confidence matches accuracy; the KD temperature is chosen so the student

learns better. A teacher distilled at large T is not thereby calibrated, and a calibrated teacher is not necessarily a good one to distil from.

2.3 Benchmarks, teacher–student pairs and metrics

Most logit-distillation papers since 2022 report results on the same three datasets, using the protocol and code base that CRD (Tian et al., 2020) introduced for CIFAR-100 and that DKD's `mdistiller` repository (Zhao et al., 2022) extended to ImageNet and COCO. *CIFAR-100* has 100 classes of 32×32 images, with 50k training and 10k validation images; *ImageNet-1K* has 1,000 classes with about 1.28M training and 50k validation images; and *MS-COCO* is used for 80-class detection with the 118k-image `train2017` and 5k-image `val2017` splits (split sizes as reported by Zhao et al. (2022), Sec. 4).

CIFAR-100 papers use homogeneous pairs (e.g. ResNet32×4→ResNet8×4, WRN-40-2→WRN-16-2) and heterogeneous ones (e.g. ResNet32×4→ShuffleNet-V1/V2); on ImageNet the standard pairs are ResNet34→ResNet18 and ResNet50→MobileNet (Zhao et al., 2022). ResNet32×4→ResNet8×4 is the most quoted: in this protocol the teacher reaches 79.42%, the student trained alone 72.50%, and classical KD 73.33% top-1 (Zhao et al., 2022, Table 6). We use this pair for Table 3.

The headline metric is top-1 accuracy (on CIFAR-100 averaged over runs, five in DKD, Zhao et al., 2022); detection reports COCO mAP (Li et al., 2023c), and newer papers add calibration error and training cost (Hossain et al., 2025b).

Two cautions apply to every table in this survey. First, papers do not reuse each other's numbers consistently. On ResNet32×4→ResNet8×4, CTKD is listed at 73.39% by Sun et al. (2024) but at 73.70% by Xu et al. (2025), and LSKD reports 76.62% for itself while Xu et al. (2025) list it at 76.16%. Such 0.3–0.5-point gaps exceed several differences between 2023–2025 methods in Table 3. Second, Miles et al. (2024) argue that CIFAR-100 is increasingly saturated as a KD benchmark, so some reported gains may come from regularisation rather than better transfer; whether this holds for a given method is open. We therefore report each method's own number with its source table and do not rank methods whose gaps are within this spread.

3 Logit-Based Distillation

Logit-based methods supervise the student through the teacher's output layer; in their output-only form they need no projector and fit any pair of classifiers with the same label space. Until 2022 they trailed feature methods: on ResNet32×4→ResNet8×4, classical KD (Hinton et al., 2015) reaches 73.33% against 75.51% for CRD and 75.63% for ReviewKD (Zhao et al., 2022, Table 6). Later work closed the gap by changing what the KL term of Eq. (2) measures. We group methods by how they decompose the loss (§3.1), rescale the logits (§3.2), or replace the divergence (§3.3); each method's own number on the common pair is in §9 or in supplementary Table S2.

3.1 Decoupling the KD loss

DKD (Zhao et al., 2022) splits KL into a binary target-class term (TCKD) and a non-target term (NCKD). In classical KD the non-target term is scaled by $1 - p_y^T(\mathbf{z}_t)$ and so vanishes on the confidently predicted samples that DKD's ablation finds most useful; DKD removes this coupling with separate weights, $\alpha\,\mathrm{TCKD} + \beta\,\mathrm{NCKD}$, and lifts ResNet8×4 from 73.33% to 76.32%, above CRD and ReviewKD at the cost of plain KD (Zhao et al., 2022, Table 6). Its code base (`mdistiller`) underlies most later logit work. The authors state its limits: β must be tuned per pair, and a logit-only loss cannot transfer the localisation knowledge detectors need, so DKD trails ReviewKD on COCO. Several later methods normalise the non-target part (NKD, Yang et al., 2023) or distil relations between local class pairs; their gains over DKD are mostly within the cross-paper spread of §2.3. LDRLD (Xu et al., 2025), the dense local-relation variant, reports 77.20% (Xu et al., 2025, Table 1), at a cost that grows with the number of pairs. GDKD (Zheng & Cheng, 2025) generalises the target/non-target split into a partition between the teacher's top classes and the rest, reporting 77.76%

against 76.64% for its own DKD run (Zheng & Cheng, 2025, Table I), a baseline 0.32 points above DKD's figure.

3.2 Rescaling the logits: temperature and standardisation

CTKD (Li et al., 2023c) learns the temperature adversarially through a gradient-reversal layer on a curriculum; gains are modest but consistent (ImageNet KD 70.83%→71.32%, DKD 71.13%→71.51%, Li et al., 2023c, Table 4), and LSKD's reproduction gives KD+CTKD 73.39% on our pair (Sun et al., 2024, Table 2). LSKD (Sun et al., 2024) argues that a small student need not reach the teacher's logit range, since only relations between logits matter, and z-score standardises each logit vector before the softmax, a per-sample temperature. As a plug-in it lifts KD from 73.33% to 76.62%, DKD to 77.01% and MLKD from 77.08% to 78.28% (Sun et al., 2024, Table 2), so standardisation and decoupling combine; gains shrink on other pairs (0.13 points for DKD on VGG13→VGG8) and LDRLD reproduces KD+LS at 76.16% (Xu et al., 2025, Table 1). LumiNet (Hossain et al., 2025b) normalises each class logit with batch statistics instead, reporting 77.50% over five runs and an expected calibration error of 0.06 against 0.11 for KD (Hossain et al., 2025b, Tables 1 and 5); its targets therefore depend on batch composition.

3.3 Replacing or enriching the KL divergence

DIST (Huang et al., 2022) targets the capacity gap by replacing KL with inter- and intra-class Pearson correlation, which keeps the ranking of predictions while ignoring shift and scale (76.31%, ImageNet 72.07%, Huang et al., 2022, Tables 5 and 2); it is weaker for binary tasks. MLKD (Jin et al., 2023) aligns predictions at instance, batch and class level and augments them over several temperatures, reporting 77.08% (Jin et al., 2023, Table 2); part of its gain may come from the temperature augmentation. WKD replaces KL with a Wasserstein distance that uses relations between classes (Lv et al., 2024).

Batch and sample structure is the last axis. SKD (Jeong et al., 2026) adds batch Gram-matrix alignment to instance KL and reports 78.33%, the highest single-method logit number we found, though its KD baseline of 74.75% is about 1.4 points above the usual 73.33% (Jeong et al., 2026, Table 1). EA-KD (Su et al., 2025) weights samples by entropy, 75.46% alone and 78.38% stacked as EA-MLD+LS (Su et al., 2025, Table 2). RLD (Sun et al., 2025) uses the label to mask every class the teacher ranks at or above the true class, so a wrong teacher ranking is not transferred (76.64%, 0.32 above DKD, Sun et al., 2025, Table 1); it and EA-KD are analysed further in §6.

3.4 Comparison and outlook

Since LSKD wraps several bases (Sun et al., 2024, Table 2), decomposition, rescaling and divergence behave as separable axes. The 2024–2025 methods report 76.0–77.5%, a range comparable to cross-paper reproduction spread (§2.3), so we name no single best method; only the step of about three points from KD to the 2022 methods is consistent across sources. DKD and KD+LSKD are cheap re-reported baselines with public code (Xu et al., 2025, Table 1); the highest values (MLKD+LSKD, SKD, EA-MLD+LS) come with extra terms, a stronger baseline, or stacking. Recent work shifts to which teacher signal to trust (RLD; §6), class imbalance, and balancing the KD term against the task loss (§8). §9 and supplementary Table S2 give the full included set.

4 Feature- and Relation-Based Distillation

Feature methods expose intermediate representations and relation methods transfer structure among examples. The design question is whether the student should reproduce teacher coordinates, a transformation of them, or only selected invariants (Fig. 4): direct matching presumes correspondence, while relaxing it suits heterogeneous architectures.

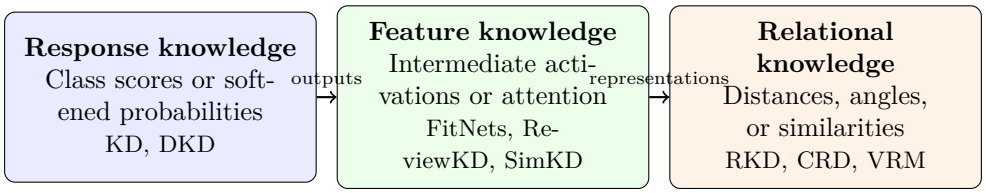


Figure 4: Knowledge targets distinguish output responses, internal representations, and structure between examples. A method may combine targets; these categories describe its principal transferred signal.

4.1 Feature alignment and projector design

FitNets guide a thin student with intermediate "hints" through a learned regressor that reconciles dimensions, giving early layers a signal output-only KD lacks; a poorly chosen layer or projection can constrain rather than help (Romero et al., 2015). ReviewKD aggregates multi-stage teacher features into review representations for each student stage, at the cost of extra feature paths (Chen et al., 2021). SimKD projects the student's penultimate feature into the teacher classifier's space and reuses the frozen teacher head, which preserves the teacher's decision boundary but must be kept at inference (78.08%) (Chen et al., 2022). The 2024 projector study (SRD) warns that a projector's gain is entangled with normalisation and loss design, so projector-only ablations should come before crediting a loss; it reports 76.44% with RandAugment (78.37% with non-standard SSKD teachers, not comparable) (Miles & Mikolajczyk, 2024).

4.2 Cross-architecture transfer

For heterogeneous pairs, the common strategy is to relax direct channel matching: supervision moves to a shared space or to invariants that do not assume stage-wise correspondence. OFA-KD projects intermediate student features into the logit space (Hao et al., 2023). Such methods often lack results on the standard homogeneous pairs.

4.3 Relational and contrastive transfer

RKD matches pairwise distances and angles between examples rather than coordinates, freeing channel correspondence at the cost of batch-dependent targets (Park et al., 2019). CRD maximises teacher–student agreement for the same example against negatives, which needs negative-sampling memory and is sensitive to the sampling protocol (Tian et al., 2020). VRM builds affinity graphs over real and virtual samples and prunes unreliable edges, so its gains depend on augmentation quality (78.76%) (Zhang et al., 2025). TopoKD distils feature-space topology through persistence diagrams (75.40%; 73.60% on ImageNet, above the 73.31% teacher it prints) (Kim et al., 2024). WKD-F applies a Wasserstein distance to Gaussian feature distributions (Lv et al., 2024).

4.4 Comparison of feature and relation methods

Three questions separate these methods: where supervision is attached (one stage, the classifier input, all stages, or a teacher tail), which sets the need for teacher activations; what invariance is requested (coordinates, attention, pairwise geometry, or low redundancy), where looser constraints cover more backbones; and what training work is added (a projector, negatives, virtual neighbours, or denoising). For same-family compression, FitNets, ReviewKD or SimKD fit when stages or the classifier are compatible. For cross-family pairs, methods that relax channel matching trade it for assumptions about the shared space. RKD, CRD and VRM make the batch or distribution the unit of supervision, so batch size, negatives and virtual samples become error sources. These are heuristics and do not rank the methods. Standard-pair results are in §9 and supplementary Table S2.

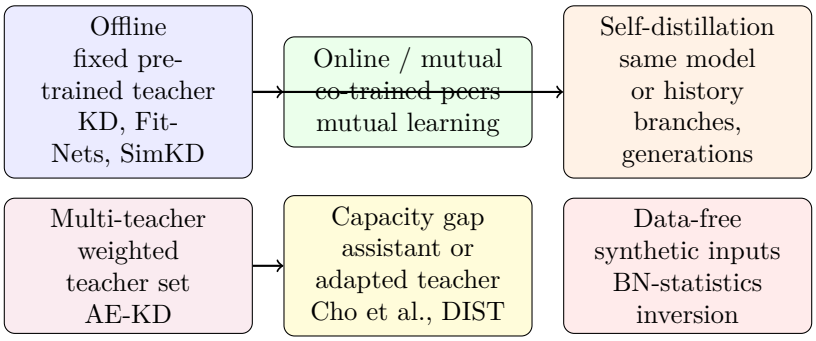


Figure 5: Training schemes differ in teacher availability, number of guidance sources, and whether original training data are available. Arrows indicate alternative arrangements, not a required progression.

5 Distillation Schemes

A training scheme fixes when the teacher exists, how many sources supervise the student, and what data the trainer sees; these choices shape compute and failure modes as much as the loss does (Fig. 5).

Offline distillation. A teacher is trained first, frozen, and then supervises the student together with the labels (Hinton et al., 2015). Feature methods keep this arrangement and change only what is transferred; SimKD, for example, reuses the frozen teacher classifier on a projected student feature (Chen et al., 2022). Offline KD is reproducible, but the frozen teacher cannot adapt to a mismatch the student exposes, and feature methods also need the teacher's internals; with output-only access, logit KD is the fitting choice.

Online and mutual distillation. Online schemes train teacher and student, or several peers, together instead of fixing a pretrained teacher. SHAKE sits between the offline and online settings: a shadow head on the frozen teacher is trained together with the student (Li & Jin, 2022). Results that depend on training both networks are not comparable with offline numbers. Online schemes suit settings with no fixed teacher, but early errors can be reinforced mutually.

Self-distillation. Targets come from the same model or its own training history rather than from a separate teacher. DLB reuses the previous mini-batch's soft predictions on overlapping samples (Shen et al., 2022), and USKD builds customised soft labels with no teacher at all (Yang et al., 2023). Self-distillation avoids a second model, but can recycle its own errors over generations.

Multi-teacher distillation. Adaptive Ensemble KD combines teachers in gradient space (Du et al., 2020). Multi-teacher schemes pay off when teachers cover complementary classes, domains or groups, at a cost that rises with teacher count.

Teacher assistants and capacity gap. Cho & Hariharan (2019) showed that a stronger teacher does not always give a better student, so teacher accuracy alone does not predict transfer. Responses to this gap either change the teacher the student sees or alter the teacher's signal. DPK feeds part of the teacher's features to the student as dynamic prior knowledge, so that a larger teacher yields a better student (Qiu et al., 2023); DIST (§3) addresses the same gap with a correlation loss (Huang et al., 2022).

Augmented teacher views and transfer data. TeKAP perturbs one teacher's features and logits into several virtual teachers (Hossain et al., 2025a). Deriving views from one teacher avoids ensemble cost, but every view inherits that teacher's errors. Changing the transfer data instead trades fidelity to the original training set for cost or availability.

Searched distillers. Auto-KD (Li et al., 2023b) and KD-Zero (Li et al., 2023a) search the distiller itself (knowledge transform, distance and location) for each teacher–student pair, by tree search and by evolution

Table 2: Reliability signals and treatment of teacher supervision. †: outside image classification (speech, text generation or summarisation); background for the mechanism, not compared with §9.

Method	Reliability signal	What happens to rejected supervision	Limitation
CTSL† (Meng et al., 2019)	Teacher predicts the observed label	Replaced by hard-label supervision	Speech adaptation; label errors affect the gate
CCKD (Mishra & Sundaram, 2021)	Teacher probability on the observed label	Downweighted or blended with one-hot targets; easy examples may be skipped	Preprint; variants have different loss behavior
Hard Gate KD† (Lee et al., 2022)	Calibrated teacher and student true-token/sentence probabilities	Routes to label supervision when the gate rejects teacher advice	Language-generation setting; no per-example CE guarantee
EA-KD (Su et al., 2025)	Teacher and student entropy	Up-weights high-entropy samples	Proxy signal; no direct validation of per-sample benefit
TPKD (Yuan & Tao, 2026)	Task-margin preservation	Projects teacher directions before transfer	2026 preprint; abstract omits teacher/student setting
LR-DS / RLD (Lan et al., 2025; Sun et al., 2025)	Label disagreement and influence / label-refined logits	Revises targets and selects examples / retains refined logits	Refinement can preserve misleading relations
MCD (Mishra et al., 2026)	Whether the teacher is correct	KD weighted down when the teacher is wrong; inter-class relations still taken	Needs labels; no extra parameters
CHAD† (Sumit et al., 2026)	Validation-gradient alignment	Learns sample weights	Unreviewed preprint; evidence is task-specific
KD-AIF (Wu et al., 2025)	Influence functions	Samples down-weighted	Also updates the teacher
SCKD (Zhu & Wang, 2021)	Similarity between KD and student-task gradients	Suppresses KD gradient when it conflicts with student objective	Gradient similarity is a local proxy

respectively. Their reported numbers come after a per-pair search, so they measure the search as much as any single loss (§8).

Data-free distillation. When training data cannot be used, the student is trained on synthetic inputs derived from the teacher. NAYER moves the generator's randomness into a noisy layer and conditions it on label-text embeddings to speed up synthesis (Tran et al., 2024). Results depend on the synthesis budget, so they are not comparable with the other tables. Synthetic coverage is the central failure mode.

No scheme dominates; the choice should follow the constraint that motivates KD (no teacher, a large capacity gap, privacy, compute, or no single reliable teacher).

6 Teacher Reliability and Selective Transfer

The methods below judge teacher reliability by different signals, summarised in Table 2. Several of the papers come from speech, text generation or summarisation; they are background for the mechanism, outside our image-classification scope, and their numbers are not compared with the tables of §9. Conditional Teacher-Student Learning (CTSL) uses teacher targets only when the teacher predicts the observed label, while Confidence Conditioned KD (CCKD, an arXiv preprint) uses the teacher's probability on that label as a sample weight or to blend teacher and one-hot targets, and its self-regulation variant can skip examples the student already handles (Meng et al., 2019; Mishra & Sundaram, 2021). CTSL makes a discrete correctness decision; CCKD keeps graded confidence. Both rely on observed labels, so label errors corrupt the signal, and correctness says nothing about calibration, student compatibility, or the value of the other class probabilities.

Hard Gate KD routes language-generation supervision by comparing the student's probability on the true token or sentence with the calibrated teacher's, switching between teacher and label (Lee et al., 2022). Unlike CTSL it uses the student's state and calibration, and the label branch may be inactive when the teacher is selected, so no per-example ground-truth loss is guaranteed. The post-hoc calibration temperature plays a different role from the KD temperature (Guo et al., 2017).

EA-KD weights examples by teacher and student entropy, up-weighting high-entropy samples without a threshold; entropy does not say which class relations to trust (Su et al., 2025). TPKD (a 2026 preprint) projects teacher directions to preserve task margins, asking whether an update threatens the task rather than whether the teacher looks uncertain, which links it to the update rules of §8 (Yuan & Tao, 2026).

Other methods repair targets or estimate usefulness. Label Revision and Data Selection (LR-DS) revises teacher labels with ground truth and selects a distillation subset by influence; examples outside it keep label supervision (Lan et al., 2025). RLD refines teacher logits with labels, keeping useful class relations when the teacher's top class is wrong, unlike CTSL's switch; it does not show that every retained relation helps, and its evidence covers CIFAR-100 and ImageNet only (Sun et al., 2025). MCD weights the KD and CE terms by whether the teacher is correct: a correct teacher gets a large distillation weight, while for a wrong teacher the student leans on the label but still takes inter-class relations from the teacher; it needs no extra parameters and reports 76.81% on ResNet32×4→ResNet8×4 (Mishra et al., 2026). CHAD learns sample weights from the alignment between distillation and validation-loss gradients; on BanSum, teacher-confidence and surface-agreement gates stayed within 0.0001 ROUGE-L of cross-entropy, while CHAD improved over standard KD by 0.0173 ROUGE-L, and the paper's multilingual result is for EWAD+CPDP, not CHAD (Sumit et al., 2026). Gradient alignment is a local proxy rather than a causal test, and the gains are specific to low-resource summarisation, so CHAD is background here.

KD-AIF weights samples with influence functions but also updates the teacher (to 86.56), so its 82.20 is not comparable with fixed-teacher results (Wu et al., 2025).

Comparing reliability signals. Label agreement and calibration do not say whether a given update helps a given student. CCKD and EA-KD weight, CTSL and Hard Gate KD route, SCKD gates by gradient similarity, RLD repairs, LR-DS selects, and CHAD learns weights. Their failure cases differ: a label gate inherits annotation errors, a confidence weight trusts confident errors, and a gradient proxy depends on the validation objective. Group-level methods should be judged with group metrics, not ranked beside sample gates on overall accuracy (Meng et al., 2019; Mishra & Sundaram, 2021; Lee et al., 2022; Su et al., 2025; Sun et al., 2025; Lan et al., 2025; Sumit et al., 2026; Zhu & Wang, 2021). As a heuristic: selection fits when harmful targets can be replaced by reliable labels, repair when useful class relations survive a wrong top prediction, group-aware methods fit systematic bias, and task-preserving projection fits interference with the student's objective. The papers span speech, generation, classification, group-robust benchmarks and summarisation, so there is no common protocol (Yuan & Tao, 2026).

7 Multi-Objective Optimization: A Primer

A distillation loss is a sum of a task loss and terms that pull the student towards the teacher. §8 uses one part of multi-objective optimisation (MOO): the first-order rules over per-objective gradients defined below. Chen et al. (2026b) survey the wider field.

Pareto optimality and stationarity. Consider K differentiable losses $L_1, \ldots, L_K$ of shared parameters θ, with gradients $g_i = \nabla_\theta L_i(\theta)$. A point $\theta^\star$ is Pareto optimal when no other point lowers one loss without raising another; gradient methods target the local notion of *Pareto stationarity*, where some convex combination of the gradients vanishes, $\sum_i \alpha_i g_i = 0$ with $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$. Every Pareto optimal point is Pareto stationary but not conversely (Sener & Koltun, 2018). Away from such points some step lowers all losses; the rules below differ in which direction they pick.

Linear scalarisation. The default in distillation is the weighted sum $L = \sum_i w_i L_i$ with fixed $w_i \geq 0$, whose gradient is $\sum_i w_i g_i$, at one backward pass. As the weights vary it reaches only the convex part of the Pareto front (Gong et al., 2021), and its direction follows raw gradient norms, so a much larger gradient dominates unless the weights are retuned. Whether this matters in practice is contested (§8.2).

7.1 Symmetric rules

The first family, from multi-task learning, treats all objectives alike.

MGDA. Sener & Koltun (2018) cast multi-task learning as MOO with the multiple-gradient descent algorithm (MGDA), which picks the minimum-norm point of the convex hull of the gradients, $\min_{\alpha\in\Delta} \| \sum_i \alpha_i g_i\|_2^2$. For two objectives the weight has a closed form, $\hat{\alpha} = \left[(g_2 - g_1)^\top g_2 / \|g_1 - g_2\|^2\right]_{[0,1]}$; for more objectives, Frank–Wolfe; MGDA-UB takes gradients at a shared representation to need one backward pass (Sener & Koltun, 2018). Two-objective MGDA favours the *smaller* gradient (an interior minimiser has equal inner products with both), so with a large task and a small distillation gradient it moves mostly along distillation (Fig. 6).

PCGrad. Yu et al. (2020) keep each task gradient but edit it when it conflicts with another. If $g_i^\top g_j < 0$, PCGrad replaces g_i by its projection onto the normal plane of g_j, $g_i \leftarrow g_i - \frac{g_i^\top g_j}{\|g_j\|^2} g_j$. It repeats this for every other task in random order and sums the results. Unlike MGDA it leaves non-conflicting gradients untouched and keeps the scale of the sum; it needs one backward pass per objective and gives no priority, so with two objectives the task gradient loses its distillation component as much as the reverse.

CAGrad. Liu et al. (2021) seek, within a ball around the average gradient g_0, the direction that most improves the worst-off objective: $\max_d \min_i \langle g_i, d\rangle$ subject to $\|d - g_0\| \le c\|g_0\|$. The dual is a problem over the simplex, $\min_{w\in\Delta} g_w^\top g_0 + \sqrt{\phi}\|g_w\|$ with $\phi = c^2\|g_0\|^2$, and the update is $d = g_0 + \frac{\sqrt{\phi}}{\|g_w\|} g_w$. The radius c interpolates between gradient descent on the average loss ($c = 0$) and MGDA (large c); CAGrad inherits MGDA's symmetry and per-objective backward passes and adds one hyperparameter.

Other symmetric rules. Other symmetric rules solve a scale-invariant bargaining game, amortise MGDA to one backward pass, or retune scalar loss weights, which addresses scale but not conflict.

7.2 Priority, lexicographic and constrained rules

The second family makes one objective primary and the others constraints, matching the usual reading of distillation: the labels must be fit and the teacher is useful but fallible. Each rule finds the direction closest to a reference gradient subject to linear inequalities on the others. DBGD minimises $\|\nabla f - v\|^2$ subject to $\nabla g^\top v \ge \phi(\theta)$, with closed form $v = \nabla f + \lambda \nabla g$ and a barrier that forces progress on g until its target is met; the authors argue a tuned weighted sum cannot solve the constrained problem when the duality gap is non-zero (Gong et al., 2021). Continual-learning methods project the new-task gradient so that earlier-task gradients do not get worse, i.e. with zero required progress. PNGD stays close to an external criterion ∇F with one scheduled margin on all losses (Ye & Liu, 2022), and AuxiNash learns a preference vector over a Nash bargaining solution, so its priority is learned and the main task's progress is not guaranteed (Shamsian et al., 2023).

PCD. Priority-constrained descent (Varam & AlHajri, 2026) rescales each gradient by an EMA of its norm and solves $\min_{\tilde{d}} \frac{1}{2}\|\tilde{d} - \tilde{g}_1\|^2$ subject to $\tilde{g}_j^\top \tilde{d} \ge \tau \|\tilde{g}_j\|^2$; for two objectives, $\tilde{d}^\star = \tilde{g}_1 + \max\left(0, \tau - \tilde{g}_1^\top \tilde{g}_2 / \|\tilde{g}_2\|^2\right)\tilde{g}_2$, rescaled to the primary gradient norm, with no correction when the constraint already holds. The authors say the program is not new (DBGD at $K = 2$; continual-learning gradient projection at $\tau = 0$); they add the normalisation, which makes τ scale-free, and a general-K analysis. Each secondary makes first-order progress, but the primary is not guaranteed to descend under strong conflict, no convergence theorem covers the deployed algorithm, and, unlike AuxiNash, the priority is fixed rather than learned. Its evidence comes from pruning and low-rank compression, not distillation.

7.3 The two families side by side

In Fig. 6, MGDA (83°) leans to the shorter distillation gradient; CAGrad sits between the average and MGDA (66°); PCGrad returns the longest vector (53°); PCD (46°) is the closest to the task among the conflict-aware rules; the equal-weight sum (41°) is closest only because the task gradient is longer. Symmetric rules suit settings where every objective must improve, priority rules settings where one objective is the reason for training; whether any beats a tuned weighted sum is the empirical question of §8.2.

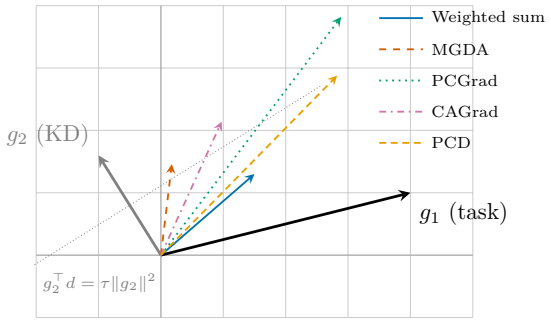


Figure 6: Directions chosen by five rules for one pair of conflicting gradients, $g_1 = (2, 0.5)$ (task) and $g_2 = (-0.5, 0.8)$ (distillation), $g_1^\top g_2 = -0.6$. Weighted sum with equal weights; MGDA two-vector min-norm point (Sener & Koltun, 2018); PCGrad (Yu et al., 2020); CAGrad with $c = 0.5$ (Liu et al., 2021); PCD with $\tau = 0.5$ and unit normalisation scales, so its direction lies on the dotted line $g_2^\top d = \tau\|g_2\|^2$ (Varam & AlHajri, 2026). Arrows are update directions before any rescaling. The vectors are illustrative, not measured. Generated by `figures/moo_geometry.py`; a unit test recomputes each direction.

8 Knowledge Distillation as Multi-Objective Optimization

Hinton-style distillation minimises $(1-\lambda)L_{\text{CE}} + \lambda L_{\text{KD}}$ (Hinton et al., 2015), which is the linear scalarisation of Section 7 with $K = 2$. Two loss gradients conflict when their inner product is negative (Yu et al., 2020; Zhu & Wang, 2021). Conflict is a property of the optimisation state, not of the teacher: a correct, confident teacher can still conflict with the label gradient. At temperature 1, with teacher probabilities $(0.85, 0.10, 0.05)$, student probabilities $(0.95, 0.03, 0.02)$ and label class 1, the cross-entropy logit gradient $p_s - y$ and the forward-KL logit gradient $p_s - p_t$ have inner product -0.0077. This is a local calculation in logit space, with no experiment behind it and no claim about parameter space. Whether the teacher is right is the separate question of Section 6.

8.1 How state-of-the-art distillation uses MOO

The methods below fall into three groups by update rule: symmetric balancing (MGDA line), label-primary gating and projection, and optimiser-side rules.

AE-KD: MGDA over teachers (lineage). AE-KD (Du et al., 2020) framed ensemble distillation in gradient space: it solves a soft-margin MGDA over the per-teacher KD losses (dual box constraints $0 \le \alpha_m \le C$, $C \in [1/M, 1]$; $C = 1$ is MGDA), on student-logit gradients, so it is cheap. CE sits outside the MOO step, so it weighs teachers against each other, not task against teacher.

DTO-KD: closed-form min-norm on log losses. DTO-KD (Hayder et al., 2026) maximises the worse of the two relative improvement rates; the dual picks $\pi^\star = \arg\min_\pi \frac{1}{2}\|J\pi\|^2$ with $J = [\nabla \log L_{\text{distill}}, \nabla \log L_{\text{task}}]$, with closed form $\pi_1^\star = (g_{22} - g_{12})/(g_{11} + g_{22} - 2g_{12})$ and $\pi_2^\star = (g_{11} - g_{12})/(g_{11} + g_{22} - 2g_{12})$. By Corollary 3.3 the direction has equal inner product with both gradients, so neither loss dominates; this is the two-vector MGDA rule of Section 7.1 applied to log-loss gradients. One difference is that Eq. (10) imposes only $\pi_1 + \pi_2 = 1$ while Eq. (9) asks for the simplex, so the closed form returns a negative weight whenever $g_{12} > g_{22}$ or $g_{12} > g_{11}$, the case MGDA handles by clipping. The deployed algorithm amortises π with a gradient step and softmax renormalisation and adds a feature projection and separate heads, so it is not the closed form, and a plain-logit adaptation would need its own specification. On CIFAR-100, ResNet32×4→ResNet8×4, DTO-KD reports 76.40% against 75.44% for DKD and 73.33% for KD in the same table. It also reports COCO gains with ViDT students. Its convergence claim is borrowed from earlier multi-task work, and no code is linked.

SCKD: gradient-similarity gating (lineage). SCKD (Zhu & Wang, 2021) keeps the supervised loss ungated and switches each distillation term on or off by the cosine between its gradient and the supervised

gradient: the label is primary, as in Section 7.2, but the update is the plain sum of the surviving losses. The published rule appears inconsistent with the paper's own explanation: Section 3.4 and Algorithm 1 remove a term when its cosine is *greater than or equal to* the threshold φ, with $\varphi = 0$ by default, while the surrounding text speaks of removing conflicting, that is negatively aligned, terms. Whether a removed term can return and the parameter subset for the cosine are not stated, and we found no authoritative code. SCKD costs one extra backward pass per distillation term, and gradient similarity does not show that the teacher is reliable.

SCOPE: classification-priority conflict removal. SCOPE (Chen et al., 2026a) combines two components. Selective Relation Alignment weights pairwise teacher relations by entropy-derived confidence. Gradient Conflict Resolution removes the parts of the distillation gradient that conflict with the classification gradient. Classification is the unmodified reference, so SCOPE is a one-sided projection, the asymmetric counterpart of PCGrad; unlike SCKD it edits the conflicting component instead of dropping the whole term. Its full system adds auxiliary priors and relational losses; it appeared in the Journal of Imaging and is evaluated on medical image classification, outside our benchmark scope, so we discuss only its update rule.

TPKD: preserving task margins (preprint). TPKD (Yuan & Tao, 2026) (preprint) proves that full imitation can force a student to give back correct-class margin it already has. It splits the teacher signal into a label part and a conditional part (relative probabilities among wrong classes), keeps the label gradient intact and projects the conditional gradient onto a cone of output-space directions that do not shrink any correct-class margin, at $O(K \log K)$ per example and no extra backward pass. Unlike SCOPE, TPKD protects an output-space quantity, and the guarantee transfers to parameter updates only under the paper's conditions. With a VOLO-D2 teacher and a pretrained PiT-B student on CIFAR-100 (60 epochs), TPKD reaches 88.05% against 87.58% for KD and 87.81% for DTO-KD over three seeds. The projection's own contribution is small (87.83% without it; on CLINC150 93.83% without vs 93.81% with), so the benefit comes mostly from learning the conditional part.

PCD as a reference priority rule. PCD (Varam & AlHajri, 2026), the normalised priority rule of Section 7.2, has not been evaluated on distillation and enters only as a reference point. As a KD rule it would keep the label gradient unless the teacher gradient's progress fell below τ times its squared norm, then add the smallest teacher component restoring it, which places it between SCKD (drop the term) and SCOPE (remove the conflicting part). Priority means proximity to the label gradient, not guaranteed label-loss descent, and its anchored projection has antecedents in DBGD, PNGD and AuxiNash (Gong et al., 2021; Ye & Liu, 2022; Shamsian et al., 2023).

DOT and DeepKD: separate momentum. DOT (Zhao et al., 2023) changes the optimiser: separate momentum buffers for the CE gradient ($\mu - \Delta$) and the distillation gradient ($\mu + \Delta$) let distillation dominate the trajectory, an asymmetric rule favouring the teacher, the reverse of SCKD and PCD, at one backward pass per loss. On ResNet32×4→ResNet8×4 it lifts KD from 73.33% to 75.12%, against 73.66% for the best single momentum and 73.56% for the best fixed task weight. DeepKD (Huang et al., 2025a) uses three buffers (task, target-class, non-target-class) with coefficients from each component's gradient signal-to-noise ratio, plus a dynamic top-k mask on non-target logits, and reports 77.03% for KD+DeepKD on the same pair; its table also has DKD+DOT at 76.03%, below DKD alone at 76.32%, so DOT's gain did not reproduce in DeepKD's runs.

MEMA: history-aware per-sample damping (preprint). SPOFA (Yang et al., 2026) (under review), for heterogeneous architectures, adds a momentum-driven EMA (MEMA) scaler that compares the per-sample logit-space cosine $S_{t,j}$ of task and distillation gradients with an EMA baseline B_{t-1}, weighting distillation by $\mathrm{clamp}(1 - \kappa \max(0, B_{t-1} - S_{t,j}), \omega_{\min}, \omega_{\max})$: a softer SCKD gate with no extra backward pass. On CIFAR-100 with a Swin-T teacher and ResNet18 student, MEMA alone raises the OFA-KD (Hao et al., 2023) baseline from 80.54% to 81.80%, and the full method reaches 82.39%.

CHAD: sample-level usefulness (preprint). CHAD (Sumit et al., 2026) (§6) estimates usefulness rather than teacher confidence: on BanSum, standard KD gains only +0.0003 ROUGE-L over cross-entropy and 51.3% of samples are estimated to raise validation loss.

Learned weighting and distiller search. DistPro (Deng et al., 2022) meta-learns a time-varying weight schedule over a set of KD pathways. Its outer-loop weights cannot react to conflict at a single step; it reports 77.54 on ResNet32×4→ShuffleNetV2 and 71.89 on ImageNet ResNet-34→ResNet-18. Distiller-search methods search the loss design per pair, by Monte Carlo tree search in Auto-KD (Li et al., 2023b) and by evolution in KD-Zero (Li et al., 2023a), excluding search cost from their reported numbers; they sit nearer the tuned-weight baselines of Section 8.2 than the per-step rules.

8.2 A critical view

These methods assume that task and distillation gradients conflict often enough for the rule to matter; few papers test this.

Symmetric versus priority rules. AE-KD and DTO-KD balance the losses; SCKD, SCOPE, TPKD, MEMA and PCD protect the label; DOT and DeepKD let distillation lead. Both priority camps report CIFAR-100 gains and no paper compares them under one protocol.

The scalarisation debate. Three large multi-task studies found that specialised optimisers rarely beat a tuned weighted sum. Xin et al. (2022) report, from large-scale language and vision experiments, that these methods "do not yield any performance improvements beyond what is achievable via traditional optimization approaches". Kurin et al. (2022) show that unitary scalarisation with standard regularisation matches or improves on complex multi-task optimisers, and they interpret several of those optimisers as forms of regularisation. Royer et al. (2023) revisit it at larger scale and propose population-based search over the weights instead of new optimisers. None of these covers distillation. Among the KD papers above, only DOT compares with a sweep of fixed task weights (all below DOT); the others we read use published baselines at default weights.

Scale or conflict? Some gains may come from normalisation and scale rather than conflict resolution. DTO-KD's log-loss gradients $\nabla L_i/L_i$ already remove each loss's scale, and PCD's authors say so for their largest pruning gains: regulariser gradients were one to two orders of magnitude larger than the task gradient and the task–regulariser cosine never exceeded 1.2×10^{-3}, so "the cause is objective scale rather than gradient conflict" (Varam & AlHajri, 2026). They also note that their weighted-sum grid floored the effective weight far above its best range, which overstates the gap. TPKD's ablation points the same way: removing the projection costs 0.22 points on CIFAR-100 and nothing on CLINC150. MEMA is the counter-example: in SPOFA's ablation the conflict-damping scaler alone adds 1.26 points, more than the LayerNorm projector's 0.65, so conflict handling can matter in heterogeneous distillation.

Is there measured conflict in distillation? We found one direct measurement of CE–KD conflict in parameter space. DHO (Kang et al., 2026) trains a ResNet-18 student from a zero-shot CLIP ResNet-50 teacher with 16 labelled examples per class, a few-shot setting outside our scope that we use as background. With a single head, the final-epoch cosine between the two gradients, averaged over eight datasets, is -0.39 at the first backbone stage, -0.45 at the last, and -0.76 at the classifier. We found no such measurement on the standard full-label CIFAR-100 or ImageNet benchmarks. Until one exists, claims that a distillation method works *because* it resolves gradient conflict are hypotheses, and a tuned or normalised weighted sum is the baseline every such method should beat.

9 Benchmarks and Reported Results

The numbers below are the results each included paper reports for itself, so they do not form a controlled comparison. Supplementary Table S2 lists all 112 included methods with their results; the tables here keep the top results and the baselines discussed in the text.

9.1 Protocol and comparability

Almost every included paper uses the CIFAR-100 protocol of CRD (Tian et al., 2020), as packaged in DKD's `mdistiller` code base (Zhao et al., 2022): CRD's pre-trained teachers, 240 epochs, SGD, and top-1 accuracy on the 10k validation images (§2.3). We rank by the most reported pair, ResNet32×4→ResNet8×4, where classical KD reaches 73.33% (Zhao et al., 2022, Table 6). WRN-40-2→WRN-16-2 is the second homogeneous pair and the only one most data-free papers report. On ImageNet-1K the standard pairs are ResNet34→ResNet18 and ResNet50→MobileNetV1.

Three things limit cross-paper comparison even under this protocol. First, papers report different statistics: the mean of three or five runs, the best run, or a single run. Gaps under about 0.3 points on CIFAR-100 are within the usual seed spread and should not be read as a ranking. Second, many top numbers are *stacked*: a plug-in is added to an already strong method such as MLKD (Jin et al., 2023), SimKD (Chen et al., 2022) or CRLD, the base of DeepKD's best result (Huang et al., 2025a, Table 1), so the number measures the combination. The Flags column names the base and, where given, the stand-alone number. Third, some results come from a changed setting: a retrained or updated teacher (KD-AIF, Wu et al., 2025), extra data or augmentation (SRD, Miles & Mikolajczyk, 2024), label smoothing on the student (LSKD and EA-KD), online or multi-student training, a searched distiller, or a stronger KD baseline than `mdistiller`'s, as in SKD (Jeong et al., 2026) (KD 74.75). Following our screening rules, these results are kept and flagged.

9.2 CIFAR-100 and ImageNet-1K

Seventy-two methods report a CIFAR-100 number on a standard homogeneous pair (Table 3). KD-AIF's 82.20% is listed but not ranked, because its teacher is updated to 86.56%. The five highest configurations among the rest are DeepKD at 79.25% (Huang et al., 2025a), VRM at 78.76% (Zhang et al., 2025), EA-KD at 78.38% (Su et al., 2025), SKD at 78.33% (Jeong et al., 2026) and LSKD at 78.28% (Sun et al., 2024). Only VRM and SKD are single methods here: DeepKD is stacked on CRLD, EA-KD's best result comes from EA-MLD with label smoothing (EA-KD alone gives 75.46%), and LSKD's is MLKD with logit standardisation. Among single methods with an official repository, the best are SimKD at 78.08% (Chen et al., 2022), GDKD at 77.76% (Zheng & Cheng, 2025) and LumiNet at 77.50% (Hossain et al., 2025b). Few methods report WRN-40-2→WRN-16-2; among them, DistPro reaches 76.36% (Deng et al., 2022) and DeepKD 78.42%.

On ImageNet ResNet34→ResNet18 (79 methods, Table 4) the spread is narrow: all but seven of them fall between 71.0% and 73.0%, against 71.70% for DKD. The highest values carry caveats. TopoKD reports 73.60% (Kim et al., 2024), above the 73.31% of its own teacher. Only three included papers (plus DeepKD, re-read from the camera-ready version) were read for ResNet50→MobileNetV1: MCD 73.65% (Mishra et al., 2026), DIST+ 73.47% (Huang et al., 2025b) and LB 73.25% (Lin et al., 2026) (last column of Table 4). DeepKD stacked on CRLD reports 73.34% and 74.85% on the two ImageNet pairs.

9.3 Data-free and untabulated methods

Ten data-free methods (§5) report WRN-40-2→WRN-16-2 values under their own synthesis budgets, and four more methods meet the criteria only on a pair not tabulated here; all are in supplementary Table S2 and not ranked.

9.4 Comparison across families

Logit methods. These are the most numerous and the cheapest methods. Most single logit methods fall between 76.0% and 77.8% (GDKD), the stacked results above 78% mostly combine two of them, and on ImageNet they sit within about one point of DKD.

Feature and relation methods. VRM and SimKD give the highest single-method CIFAR numbers, but SimKD reuses the teacher's classifier, so the deployed student is larger than ResNet8×4 alone (Chen et al., 2022), and VRM has no public code. Feature methods need a projector per pair and intermediate-layer access; several recent ones report only on ImageNet.

Table 3: CIFAR-100 ResNet32×4→ResNet8×4 top-1 (%): the highest reported results among the methods cited in the text, and the baselines. Each value is the paper's own claim for its best reported variant (named under Flags). Fam.: L logit/loss, F feature/relation, S training scheme, R reliability, O optimizer/combination, H hybrid. Code: the authors link an official repository (unreachable on 2026-11-08 counts as no). Reference point: KD 73.33 (Zhao et al., 2022, Table 6). All 72 CIFAR-100 results, with their source tables, are in supplementary Table S2.

Method	Fam.	Venue	R32×4→R8×4	Code	Flags
Single methods, standard setting					
VRM (Zhang et al., 2025)	F	ICCV'25	78.76	no	
SimKD (Chen et al., 2022)	F	CVPR'22	78.08	yes	reuses teacher classifier
GDKD (Zheng & Cheng, 2025)	L	arXiv'25	77.76	yes	preprint (TNNLS); own DKD 76.64
LumiNet (Hossain et al., 2025b)	L	TMLR'25	77.50	yes	
MLKD (Jin et al., 2023)	L	CVPR'23	77.08	yes	
MCD (Mishra et al., 2026)	L	TMLR'26	76.81	no	+MLLD 77.45, +LS 77.11; 3 trials
RLD (Sun et al., 2025)	R	ICCV'25	76.64	yes	
DTO-KD (Hayder et al., 2026)	O	ICLR'26	76.40	no	
DKD (Zhao et al., 2022)	L	CVPR'22	76.32	yes	
DIST (Huang et al., 2022)	L	NeurIPS'22	76.31	yes	
DIST+ (Huang et al., 2025b)	L	TPAMI'25	76.19	yes	adds feature loss to DIST
TopoKD (Kim et al., 2024)	F	ICML'24	75.40	no	repo unreachable
DOT (Zhao et al., 2023)	O	ICCV'23	75.12	yes	KD+DOT (mdistiller)
Stacked, or setting changed (teacher, augmentation, baseline)					
KD-AIF (Wu et al., 2025)	R	arXiv'25	82.20	no	teacher retrained to 86.56; not ranked
DeepKD (Huang et al., 2025a)	O	NeurIPS'25	79.25	no	CRLD+DeepKD (stacked)
EA-KD (Su et al., 2025)	L	ICCV'25	78.38	yes	EA-MLD+LS (stacked, LS); EA-KD 75.46
SKD (Jeong et al., 2026)	H	CVPR'26	78.33	no	own KD baseline 74.75
LSKD (Sun et al., 2024)	L	CVPR'24	78.28	yes	MLKD+LS (stacked)
SRD (Miles & Mikolajczyk, 2024)	F	AAAI'24	76.44	yes	SSKD teachers, RandAugment; 78.37 with SSKD rotations

Table 4: ImageNet-1K ResNet34→ResNet18 top-1 (%): the highest reported results among the methods cited in the text, and the baselines. Conventions as in Table 3. Reference point: KD 71.03 and DKD 71.70 in the `mdistiller` README (Zhao et al., 2023). R50→MBV1: ResNet50→MobileNetV1, where only four included papers were read. The 79 ResNet34→ResNet18 results in Table S2 range from 70.67 to 73.60; five lie above 73.0 and two below 71.0.

Method	Fam.	Venue	R34→R18	R50→MBV1	Code	Flags
Single methods, standard setting						
DIST+ (Huang et al., 2025b)	L	TPAMI'25	72.39	73.47	yes	
LB (Lin et al., 2026)	F	TPAMI'26	72.30	73.25	no	
GDKD (Zheng & Cheng, 2025)	L	arXiv'25	72.26	–	yes	preprint
LumiNet (Hossain et al., 2025b)	L	TMLR'25	72.16	–	yes	
MCD (Mishra et al., 2026)	L	TMLR'26	72.08	73.65	no	3 trials
DIST (Huang et al., 2022)	L	NeurIPS'22	72.07	–	yes	
DistPro (Deng et al., 2022)	O	ECCV'22	71.89	–	yes	
DOT (Zhao et al., 2023)	O	ICCV'23	71.72	–	yes	KD+DOT
DKD (Zhao et al., 2022)	L	CVPR'22	71.70	–	yes	mdistiller
SRD (Miles & Mikolajczyk, 2024)	F	AAAI'24	71.63	–	yes	
SKD (Jeong et al., 2026)	H	CVPR'26	71.13	–	no	
Stacked, or setting changed (teacher)						
TopoKD (Kim et al., 2024)	F	ICML'24	73.60	–	no	above its 73.31 teacher
DeepKD (Huang et al., 2025a)	O	NeurIPS'25	73.34	74.85	no	CRLD+DeepKD
EA-KD (Su et al., 2025)	L	ICCV'25	71.99	–	yes	EA-DKD+LS; EA-KD 71.79
CTKD (Li et al., 2023c)	L	AAAI'23	71.51	–	yes	DKD+CTKD; KD+CTKD 71.32

Training schemes and reliability. Online and multi-student schemes score lower than offline methods but avoid a fixed pre-trained teacher; searched distillers reach 76.6–77.9% at the price of a search per pair. Reliability methods such as RLD and the teacher-retraining ones give 75.5–76.6% as single methods where they report it.

Optimizer-side methods. DOT (KD+DOT 75.12%, Zhao et al., 2023), DeepKD (79.25% stacked on CRLD, Huang et al., 2025a), DTO-KD (76.40%, Hayder et al., 2026) and DistPro (Deng et al., 2022) change

how the task and distillation gradients are combined (§8); none gates the distillation term per sample. PCD (Varam & AlHajri, 2026) has no reported number on these pairs.

A comparison set for combination rules. For a method that changes how the task and distillation terms are combined, the tables suggest three groups of baselines. Combination rules: a tuned weighted sum (Hinton et al., 2015), MGDA (Sener & Koltun, 2018), PCGrad (Yu et al., 2020), DOT (Zhao et al., 2023) and DTO-KD (Hayder et al., 2026), with DeepKD (Huang et al., 2025a) quoted from its paper since no code is linked. Loss-side methods with code: DKD, MLKD, LSKD (MLKD+LS) and GDKD, plus SimKD where the student can keep the teacher's classifier. Reliability methods that act on a wrong teacher: RLD and MCD. Since an optimiser change stacks on a loss change, each combination rule should be run on top of both KD and a strong logit loss. VRM and SKD link no code, so they are quoted with the protocol caveats above.

Applicability. Without intermediate-layer access, a logit method with public code (DKD, MLKD, LSKD, LumiNet) is the safe start; for accessible homogeneous pairs, SimKD or a projector-based feature method is stronger on the reported numbers; reliability methods fit a partly wrong teacher, and online or data-free schemes fit when a pre-trained teacher is the constraint.

10 Open Challenges and Future Directions

Each open question below is followed by the evidence for it and a direction that could test it.

A comparable leaderboard. The tables are not a controlled comparison (Section 9.1). On ResNet32×4→ResNet8×4, classical KD is 73.33% in DKD's table (Zhao et al., 2022), and 74.75% in SKD's (Jeong et al., 2026), a spread wider than the gaps between most single logit methods in Table 3, while gaps of 0.3–0.5 points sit inside the seed spread (Section 2.3). Stacking adds a second problem. Three of the five highest CIFAR-100 results are stacked (DeepKD on CRLD, EA-MLD with label smoothing, MLKD with logit standardisation; Huang et al., 2025a; Su et al., 2025; Sun et al., 2024), and several plug-ins lose much of their gain alone: EA-KD gives 75.46% against 78.38% stacked. We found no included paper with a full grid of plug-ins against bases. *Direction:* re-run the methods with public code in one code base, with fixed teachers and five seeds; check by rank correlation whether the published order survives, and train every pairing of a few bases (KD, DKD, MLKD) and plug-ins (logit standardisation, DOT, a reliability weight).

Which teacher signal to trust. The reliability methods disagree about what reliability means: label agreement, label probability, entropy or validation-gradient alignment (Meng et al., 2019; Mishra & Sundaram, 2021; Su et al., 2025; Sumit et al., 2026). Those that report on ResNet8×4 land between 75.5% and 76.6% (Section 9.4). We found no paper that tests whether these signals pick the same samples. *Direction:* on one fixed teacher, measure the overlap of the samples each signal selects and which best predicts the samples whose removal helps the student.

How to combine the objectives, and whether they conflict. The optimiser-side methods split into label-first, distillation-first and balancing camps (§8.2), no paper compares them under one protocol, and only DOT compares with a sweep of fixed task weights (Zhao et al., 2023), while three multi-task studies found tuned scalarisation hard to beat (Xin et al., 2022; Kurin et al., 2022; Royer et al., 2023). PCD (Varam & AlHajri, 2026) has no reported distillation result, so it does not yet answer the question. The only direct measurement of conflict we found is DHO's, with a zero-shot CLIP teacher and 16 labelled examples per class, where the final-epoch cosine at the classifier is -0.76 (Kang et al., 2026); for the full-label protocol with a trained CNN teacher we found none. Scale rather than conflict may explain some gains (PCD's pruning results, TPKD's 0.22-point projection ablation; Varam & AlHajri, 2026; Yuan & Tao, 2026). *Direction:* run one rule per camp on KD and a strong logit loss against a tuned weighted sum, logging the CE–KD cosine and norm ratio, and compare each rule with its implied rescaling alone.

Data-free distillation and the ImageNet gap. The data-free methods report on WRN-40-2→WRN-16-2 only, with differing budgets and some results mixing collected and generated data, so they share no

pair with the main ranking. On ImageNet, most ResNet34→ResNet18 results fall between 71.0% and 73.0%, close to DKD's 71.70% (Zhao et al., 2023), so CIFAR-100 gains of two or more points mostly shrink to under one. *Direction:* report data-free and data-based methods on the same pairs and teacher, and the ratio of ImageNet to CIFAR-100 gain over KD for the top methods.

Metrics and settings beyond top-1 accuracy. Almost every table here ranks by top-1 accuracy, and the KD temperature is distinct from calibration (Section 2.2, Guo et al., 2017), yet the only expected calibration error quoted in Section 3 is LumiNet's, 0.06 against 0.11 for KD (Hossain et al., 2025b). Reliability methods are judged by accuracy alone, without a check of whether teacher errors transfer. Training cost is rarely reported, though DOT needs two backward passes per step and searched distillers a search per pair (Zhao et al., 2023). Neighbouring settings are outside this survey's scope but raise the same questions: LLM, vision–language, adversarially robust and federated distillation; with a zero-shot CLIP teacher, DHO removes CE–KD conflict with separate heads (Kang et al., 2026). *Direction:* in the re-run, also report calibration error, corrupted-test accuracy, agreement with the teacher's errors and time per epoch.

11 Conclusion

This survey reviewed 112 knowledge distillation methods for image classification from 2022 to 2026, selected under inclusion criteria set in advance.

The methods fall into five families. Logit methods are the largest and cheapest group; feature and relation methods give some of the highest single-method CIFAR-100 numbers but need intermediate layers, and the best can enlarge the deployed student (Chen et al., 2022); training schemes trade accuracy for removing a fixed teacher; reliability methods decide which teacher outputs to trust. A smaller group changes how the task and distillation gradients are combined (Zhao et al., 2023; Huang et al., 2025a; Hayder et al., 2026). We placed it next to the general multi-objective rules, with priority-based rules such as PCD (Varam & AlHajri, 2026) as one option without reported numbers on the standard pairs.

On CIFAR-100 ResNet32×4→ResNet8×4 the best reported configurations are about six points above classical KD, yet most single methods lie within two points of one another, and on ImageNet nearly all results sit within two points. Many of the highest results stack methods, change the teacher, or add data, so gaps of a few tenths of a point say little. Optimizer-side changes stack on loss-side ones, so they should be tested on top of both plain KD and a strong logit loss.

Section 10 turns these caveats into five testable problems: a controlled re-run of the leaderboard, the overlap of teacher-trust signals, combination rules against a tuned weighted sum (Xin et al., 2022; Kurin et al., 2022; Royer et al., 2023) given that conflict has been measured only outside the standard protocol (Kang et al., 2026), the data-free and ImageNet gaps, and metrics beyond top-1.

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